CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Pharm. (New) Examination

University Theory Examination April 2016

PH119 Pharmaceutical Chemistry-II

Date: 30.04.16, Saturday **Time: 10:00 a.m. to 01:00 p.m.** **Maximum Marks: 80**

Instructions:

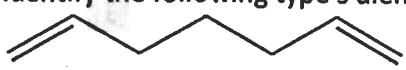
1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

Section - I

Q 1 **Attempt all questions. Each question is of one mark.** **20**

1. Following reactive intermediates are act as an Electrophile. Except One
   [A] Carbocation
   [B] Nitrene
   [C] Carbene
   [D] Carbanion
2. The following statements are correct for carbocation; Except One.
   [A] Stability order of carbocation is $3^0>2^0>1^0$
   [B] Electron donating group increase the stability of carbocation
   [C] Carbocation can act as an electrophile
   [D] Electron withdrawing group increase the stability of carbocation
3. Unit of Dipole Moment is: ______________
   [A] Cm^{-1}
   [B] Poise
   [C] Debye
   [D] ppm
4. In o- nitro phenol solution the molecules are attached by
   [A] Intermolecular H- bond
   [B] Intramolecular H- bond
   [C] Sigma bond
   [D] Pi bond

5. In the molecule of Acetylene, what is the hybridization?
   [A] SP^3
   [B] SP^2
   [C] Unhybrid
   [D] SP
6. Alkyl halides undergo a type of reaction
   [A] Nucleophilic substitution
   [B] Nucleophilic addition
   [C] Elimination
   [D] Both [A] & [C]
7. Which of the following has the highest electronegativity?
   [A] S
   [B] P
   [C] F
   [D] C
8. ____________ Intermediates are generated in Hoffmann Degradation Reaction.
   [A] Nitrene
   [B] Isocynate
   [C] Carbene
   [D] [A] and [B]
9. 2, 2, 3, 3- Tetramethylbutane is the isomer of
   [A] Octane
   [B] Heptane
   [C] Hexane
   [D] Nonane
10. Which is the example of Homogenous Catalyst?
   [A] Pd
   [B] Ni
   [C] Lindler
   [D] Wilkinson's
11. Identify the following type's dienes.



   [A] Conjugated Diene
   [B] Isolated Diene
   [C] Cumulated Diene
   [D] None of above

12. Which of the following is correct order for good leaving group in SN_1 and SN_2 reactions?
   [A] $-I > -OH > -NH_2 > -F$
   [B] $-I > -F > -NH_2 > -OH$
   [C] $-I > -F > -OH > -NH_2$
   [D] $-I > -NH_2 > -OH > -F$

13. Which of the following is the reagent for preparation of alkyl halide from alkene

[A] $SOCl_2$
[B] PBr_3
[C] Cu in ether
[D] HBr

14. All statement is correct for SN_2 reaction. Except one

[A] Follow Second order kinetic
[B] Rearrangement is possible
[C] Inversion of configuration takes place
[D] Single step reaction

15. Orientation of Elimination reaction follows...

[A] Markoniov's rule
[B] Saytzeff rule
[C] Micheal addition
[D] Hoffmann rule

16. For which mechanisms the first step involved is the same?

[A] $E_1 + E_2$
[B] $E_2 + SN_2$
[C] E_1 and SN_1
[D] SN_1 and SN_2

17. Which is not a correct about Ammonia?

[A] SP^3 Hybridization is takes place
[B] Bond angle is 107^0
[C] Shape is pyramidal
[D] Bond angle is 105^0

18. Which of the following is a polar aprotic solvent?

[A] DMF
[B] Etahanol
[C] Water
[D] All

19. Grignard reagent is reactive due to

[A] The presence of halogen atom
[B] The presence of magnesium atom
[C] The polarity of C-Mg bond
[D] All

20. Which of the following is the least acidic compound?

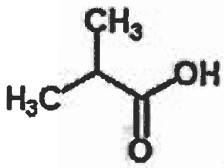
[A] CH_3COOH
[B] $ClCH_2COOH$
[C] $Cl_2CHCOOH$
[D] Cl_3CCOOH

SECTION – II

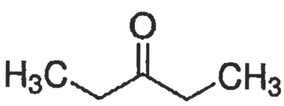
Q 2	**Attempt any <u>FOUR</u> of the following;**	**20**
A	**Explain Mechanism, Reactivity, Kinetic and Stereochemistry of SN_1 reaction.**	**05**
B	**Factor Affecting SN_1 and SN_2 reaction**	**05**
C	**Explain Mechanism, Reactivity, Kinetic and Orientation of** E_1 reaction.	**05**
D	**Explain Saytzeff's and Markonikov's Rule with examples.**	**05**
E	**Explain Stability aspects of carbocation's.**	**05**
F	**Explain Carbanion, Carbene and Free radicals in brief.**	**05**

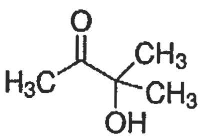
SECTION – III

Q 3	Attempt any <u>FOUR</u> of the following;	**20**
A	Write in detail the reactions of Alkenes.	05
B	Short note on (2.5 X 2 = 05 Marks) i. Hyper-conjugation ii. SP^2 Hybridization	05
C	**Elaborate in detail the free radical reaction for chlorination of Alkanes.**	05
D	**Write reaction and mechanism for Höffmann degradation of amide.**	05
E	**Explain 1, 4 addition Vs.1, 2 addition in dienes.**	05
F	**Write Any three Preparations and reactions of Alkane.**	05
		20
Q 4	**Attempt any <u>FOUR</u> of the following;**	
A	**Write any three preparations and reactions of Alkyl Halide.**	05
B	**Explain following terms; (Any Five)** I. Electrophile II. Polar Bond III. Electronegativity IV. Covalent Bond V. Inductive Effect VI. Polar Aprotic Solvent	05
C	I. **Explain Resonance Effect in details.**	03
	II. **Explain Inter and Intramolecular Forces**	02
D	Write in detail about the preparation and reaction of Alkynes.	05
E	I. **Explain Hoffmann Elimination**	02
	II. **Differentiate SP, SP^2 and SP^3 Hybridization.**	03
F	Write IUPAC Nomenclature of Following Structures.	05

1. 

2. $CH_3CH_2CH_2CH_2\overset{\overset{O}{\|}}{C}-OCH_2CH_3$

3. 

4. 

5.